

Hackathon Instructions: Analyzing Promotion Prediction Data

Objective

The goal of this hackathon is to use data analysis and machine learning to predict promotion. This means you'll determine identifying the right people for promotion, based on past data.

Hackathon site

<https://www.analyticsvidhya.com/datahack/contest/wns-analytics-hackathon-2018-1/>

Note: You need to login to access the data

In case of trouble downloading the data, try using this link [Dataset](#) to download.

Dataset

You'll be using the **Employee Performance Dataset** taken from the hackathon site. The dataset contains **employee details along with information about their performance**. The dataset includes features like:

• Employee ID	Unique identifier for each employee
• Demographics	Age, gender, region, education, etc.
• Training Details	no_of_trainings, previous_year_rating, avg_training_score, etc.
• Metrics	KPI, awards, service, etc.
• is_promoted	Target variable indicate whether the employee recommended for Promotion

Step-by-Step Instructions

- Data understanding and observations
Import Libraries, Load Data, Initial Exploration, etc.
- Exploratory Data Analysis (EDA)
Visualize Distributions, Univariate Analysis, Bivariate Analysis, etc.
- Data Cleaning
Check and Handling Missing values, Outlier Detection, Convert data types, etc.
- Feature Engineering
Encoding Categorical Data, Feature Scaling Numerical Data, Create New Features from existing columns, etc.
- Model Building
Train-Test Split, Modeling using Pipelining, Ensemble Techniques, Model Training and Prediction.
- Model Evaluation and Tuning
Evaluate Model Performance using Metrics specified in the hackathon site, Optimize model by Hyperparameter Tuning using GridSearch or others, and Compare Models.
- Evaluate Test solution
Test your solution in hackathon's website and capture result.

- **Interpret Results**
Summarize your findings, model performance, and key insights into a final report. Compare models and explain which model were most impactful in predicting promotion and choose it as final model and pickle it for using them in API and UI.
- **API Creation**
Create a FastAPI endpoint using the pickled final model. *(Take snapshot of URL links and place in your ppt slide).*
- **User Interface UI Development**
Create a Web App User Interface with the pickled final model using Streamlit library. *(Take snapshot of URL links and place in your ppt slide).*
- **Deploy API and UI both thru GCP**
Create a new repository in your personal github account and push the required files (python files, Docker files, Requirements file, pickle file, etc.) and deploy them using Google Cloud Platform (GCP) for public use *(Take snapshot of it with the respective URL links and place in your ppt slide).*
- **PowerPoint Presentation**
Prepare a PowerPoint presentation having all the steps performed along with summarizing the problem, approach, findings, recommendations and deployed links.
- **Recorded Demo**
Record a brief demo (5 minutes) walking through your code, explaining your methodology, and showcasing results.

Deliverables

Files: Submit all the Python code files, Docker files, Requirements files, Jupyter Notebook, Links of deployed API and WebApp (UI) link.

PowerPoint Presentation: Prepare a PowerPoint presentation having all the steps performed along with summarizing the problem, approach, findings, recommendations and deployed links.

Recorded Demo: Record a brief demo (5 minutes) walking through your code, explaining your methodology, and showcasing results.

Additional Notes

Documentation: Ensure your code is well-documented with comments explaining each step.

Submission: Push all the deliverables in your github and share us your github link by submitting in this Google Form [here](#)

TERMINATE/STOP ALL SERVICES/INSTANCES/BILLINGS CREATED IN CLOUD

Good luck! We look forward to seeing your innovative solutions and insights. Remember, this hackathon is not just about reaching the answer, but also about experimenting and learning along the way. Happy coding!